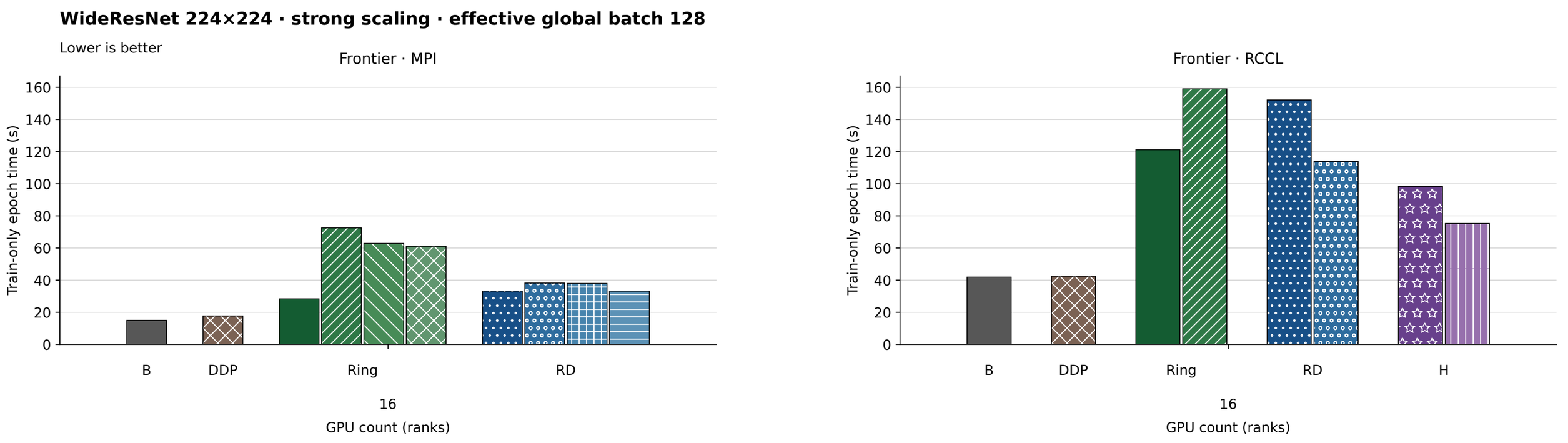

WideResNet 224×224 · strong scaling · effective global batch 128
Lower is better
Frontier · MPI
Frontier · RCCL
Train-only epoch time (s)
0
20
40
60
80
100
120
140
160
B
DDP
Ring
RD
H
16
GPU count (ranks)
Baseline
Built-in baseline
DDP diagnostics
default sync
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate